Nivaldo Humberto Oliveira Tokuda

AI & Software Engineer | Python | LLMs & RAG | Google Cloud

São Paulo, Brazil | ntokuda@utexas.edu | github.com/nivaldoh | LinkedIn

SUMMARY

Engineer with 8+ years building cloud and data systems, with recent professional experience in LLM applications, multimodal retrieval, RAG, and evaluation. Built Python AI pipelines for Paramount/CBS and Warner Bros. through Globant; delivered production systems and collaborated across disciplines at Globo. Initiated a Rust SDK for Apache Beam, later adopted and carried forward by Google engineers. MS Computer Science candidate at UT Austin, developing LLM-agent evaluation infrastructure.

PROFESSIONAL EXPERIENCE

Globant | São Paulo, Brazil

Jul 2024 - May 2026

Selected client engagements

Data Scientist | Paramount / CBS — Oct 2025 - Dec 2025

- Built multimodal video search in Python using Vertex AI embeddings, retrieving relevant news and sports video segments from text and image queries.
- Implemented video processing as multiple Cloud Run components orchestrated with Google Cloud Workflows.

Data Scientist | Warner Bros. — Jul 2025 - Sep 2025

- Built an LLM-based translation pipeline for domain-specific vocabulary using Python containers on Google Cloud Run.
- Implemented and compared retrieval-augmented generation (RAG) and model fine-tuning; used BLEU and COMET to evaluate translation quality and select the more effective approach.

Data Engineer | Google Extended Workforce (xWF) — Jul 2024 - Jul 2025

Embedded with Google teams to deliver GCP projects for enterprise customers.

- Owned the Airflow workstream of an AWS-to-GCP data warehouse migration for a US fraud-detection company. Migrated 100+ DAGs with complex business logic; supported migration of Spark jobs, BigQuery tables (including Iceberg), and Cloud Storage data.
- Developed high-throughput stock-trading data pipelines for a large US financial institution, combining transactional and analytical processing with millisecond-level latency requirements.
- Implemented Firestore backups for a Brazilian government project handling sensitive personal data, with full daily snapshots and incremental hourly backups using Cloud Functions.

Globo | São Paulo, Brazil

Jan 2022 - Jan 2024

Senior Data Engineer

- Built an internal Python application on App Engine, used by data scientists to launch Dataproc jobs. Processed user input and submitted batched requests through an internal load balancer.
- Led the creation of a data warehouse that fully replaced the previous system, reducing query times by hours and nearly halving costs. Architected data transformations and orchestrated them with Airflow deployed on Google Kubernetes Engine.
- Developed production data systems and regularly aligned technical work with cross-functional teams.

DP6 | São Paulo, Brazil

May 2017 - Dec 2021

Data Engineer

- Developed data-processing pipelines for some of Brazil's largest companies, supporting marketing analytics from user-behavior data. Collaborated with a Google team on the Megalista project using Python, Apache Beam, and Google Cloud Dataflow.

INDEPENDENT AI ENGINEERING & RESEARCH

Independent Work

May 2026 - Present

RoboCasa365 | LLM-agent and robotics evaluation (in progress)

- Integrated GR00T N1.5 and LLM-generated Python control through shared scenario-execution and evaluation infrastructure.
- Implemented current-goal scoring and separate records of scene and goal changes, information delivery, and task completion to support evaluation and failure analysis.

Total War: Warhammer III | LLM battle-control prototype (in progress)

- Built an end-to-end prototype connecting mod-exposed battle state to an LLM API and executing model-generated commands in-game.

Metamon | Collaboration with UT Austin Robot Perception and Learning Lab

- Reduced a data-processing component from 20+ hours to approximately 5 minutes by caching Pokémon battle simulation instances.
- Improved data-scraping efficiency by adding concurrency and exponential backoff.
- Implemented profiling to identify performance bottlenecks and fixed a PyTorch compiler graph break, improving GPU-based model-training speed by approximately 11%.

SELECTED OPEN-SOURCE CONTRIBUTIONS

Apache Beam | Rust SDK

2023

- Kick-started a Rust SDK for Apache Beam over three months, implementing foundational PCollection structures and synchronization primitives. The work was subsequently adopted and carried forward by Google engineers. Issue | Commits

Apache Beam | Python

2022

- Added the LogElements PTransform, unit tests, and updates to existing examples. Contribution merged into Apache Beam. PR #23879

Google Megalista | Python / Google Cloud

2021

- Implemented a Firestore-backed configuration source as an alternative to Google Sheets for clients with restricted access to spreadsheets. Contribution merged into the project. PR #12

TECHNICAL SKILLS

Applied AI: LLM APIs, RAG, model fine-tuning, multimodal embeddings, LLM/agent evaluation; Vertex AI.

Programming and ML: Python, SQL, Rust, PyTorch; model-training profiling.

Cloud/backend: Google Cloud, containerized Python services, Cloud Run, Workflows, Kubernetes (GKE), App Engine, Cloud Functions.

Data systems: BigQuery (SQL), Firestore (NoSQL), Apache Airflow, Apache Beam, Dataflow, Spark, Dataproc, Cloud Storage.

EDUCATION

University of Texas at Austin | MS, Computer Science

Aug 2024 - Dec 2027 expected

Online. Selected coursework: Parallel Systems, Deep Learning, Reinforcement Learning, NLP.

University of Illinois at Urbana-Champaign

May 2024 - Jul 2024

Master of Computer Science studies (online; transferred to UT Austin). Coursework: Theory and Practice of Data Cleaning; Programming Languages & Compilers.

Universidade Presbiteriana Mackenzie | BS, Computer Science

Jul 2017 - Jul 2022

Teaching Assistant, Computer Programming I (C++).

LANGUAGES

Portuguese: native. English: advanced (C1).